

Exercise 1: for...in Iterations

(1a) Using `range()`, write a `for` loop that generates the output line:

1 2 3 4 5 6 7 8 9 10

With a tab separation between the numbers. (Hint: use `end="\t"` in the print command)

(1b) Based on (1a), write nested `for` loops that generate as output the following multiplication table:

1	2	3	4	5	6	7	8	9	10
2	4	6	8	10	12	14	16	18	20
3	6	9	12	15	18	21	24	27	30
4	8	12	16	20	24	28	32	36	40
5	10	15	20	25	30	35	40	45	50
6	12	18	24	30	36	42	48	54	60
7	14	21	28	35	42	49	56	63	70
8	16	24	32	40	48	56	64	72	80
9	18	27	36	45	54	63	72	81	90
10	20	30	40	50	60	70	80	90	100

(Hint: use `print()` when it is needed that the output will go one line down.)

(1c) Modify your solution to (1b), so that the output will be:

1									
2	4								
3	6	9							
4	8	12	16						
5	10	15	20	25					
6	12	18	24	30	36				
7	14	21	28	35	42	49			
8	16	24	32	40	48	56	64		
9	18	27	36	45	54	63	72	81	
10	20	30	40	50	60	70	80	90	100

Exercise 2: while Iterations

Write a code, containing a `while` loop, which starts with `amount = 100` and `year = 0`, and in which in each iteration `year` is increased by 1 and `amount` is multiplied by 1.03

These iterations should proceed until `amount` becomes ≥ 120 .

In each iteration the `year` and `amount` values are printed.

The obtained output should be:

Year: 1	Amount: 103.00
Year: 2	Amount: 106.09
Year: 3	Amount: 109.27
Year: 4	Amount: 112.55
Year: 5	Amount: 115.93
Year: 6	Amount: 119.41
Year: 7	Amount: 122.99

Exercise 3: List comprehension

Using `range()`, write a list comprehension statement that generates a list of the squares (`**2`) of all the integers from 1 to 30 which can be divided by 3 (with no remainder).

The obtained results should be:

[9, 36, 81, 144, 225, 324, 441, 576, 729, 900]

Exercise 4: Functions

(4a) Write a function `convert_temperature` to convert between Celsius and Fahrenheit.

The function should take two inputs:

- `temperature` : for the temperature value
- `unit` : for the unit, "F" for Fahrenheit and "C" for Celsius.

The conversion from Celsius to Fahrenheit is: $\text{temperature} * 9/5 + 32$

The conversion from Fahrenheit to Celsius is: $(\text{temperature} - 32) * 5/9$

Check the function you wrote for the following inputs:

```
convert_temperature(0, "C")
```

```
0 Celsius are 32.0 Fahrenheit
```

```
convert_temperature(32, "F")
```

```
32 Fahrenheit are 0.0 Celsius
```

```
convert_temperature(50, "F")
```

```
50 Fahrenheit are 10.0 Celsius
```

```
convert_temperature(10, "C")
```

```
10 Celsius are 50.0 Fahrenheit
```

(4b) Write a function `factorize_integer` to factorize integers (e.g. $30 = 2 * 3 * 5$). The function should take one input `number` and output its prime factors.

Check the function you wrote, based on the following examples:

```
factorize_integer(2025)
```

```
2025 = 3 * 3 * 3 * 3 * 5 * 5  
5 is prime
```

```
factorize_integer(2026)
```

```
2026 = 2 * 1013  
1013 is prime
```

```
factorize_integer(2027)
```

```
2027 = 2027  
2027 is prime
```

```
factorize_integer(1234567890)
```

```
1234567890 = 2 * 3 * 3 * 5 * 3607 * 3803  
3803 is prime
```

Check now also for following example (12345678901234567). Here, if the algorithm your wrote is not efficient enough, the computation may take long time (more than several seconds). In that case, interrupt the Kernel (e.g. through the Jupyter Notebook menu) and try to make the function more efficient algorithm-wise.

```
factorize_integer(12345678901234567)
```

```
12345678901234567 = 7 * 1763668414462081  
1763668414462081 is prime
```